Multiple Regression Analysis with Qualitative Information

These notes accompany Lecture 5. They combine the slide material with the explanations given in class. Terms printed in bold are defined in the glossary at the end.

Until now every variable in our models has had quantitative meaning – wages, years of education, firm sales. But gender, ethnicity, industry, region and country income level are qualitative, and they matter. This lecture shows how to bring qualitative information into a regression, first as explanatory variables and then as the dependent variable.

Learning outcomes – after this lecture you should be able to:

1. Define a dummy variable and interpret its coefficient relative to the base group.
2. Explain why a model with an intercept can include only g − 1 dummies for g groups.
3. Interpret a dummy coefficient when the dependent variable is in logs, both approximately and exactly.
4. Interpret interactions between dummies, and recover the differential for every group.
5. Use a slope dummy to test whether a relationship differs across groups, and explain why centring changes the intercept dummy's meaning.
6. Estimate and interpret a linear probability model, and state its three drawbacks.
7. Explain what logit and probit fix, and why their coefficients cannot be read as probabilities.
8. Compute and distinguish the partial effect at the mean (MEM) and the average partial effect (APE).

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1. The nature of qualitative information

Qualitative factors often come in binary form: a person is female or male; a household does or does not own a computer; a country is high-income or it is not. In every such case the relevant information can be captured by a variable taking only the values zero and one.

Such a variable is a **binary variable**, or – as it is almost always called in econometrics – a **dummy variable**. The name is not especially descriptive, but it is universal.

Examples where qualitative information plausibly matters:

- Gender may play an important role in determining salaries.
- Different ethnic groups may follow different behavioural patterns.
- Educational levels can affect earnings from employment.
- Changes in a political regime may affect economic conditions.
- A war can affect all aspects of economic activity.
- A country's income level may be associated with a different reaction to shocks.

> **Why zero-one coding, specifically?**
>
> The benefit of capturing qualitative information with zero-one variables is that the resulting parameters have very natural interpretations – differences between groups, read straight off the coefficient.
>
> In defining a dummy you must decide which event gets the value one. The choice is arbitrary, but it determines the sign of the coefficient, so state it explicitly.

2. Intercept dummy variables

2.1 A single dummy independent variable

In the simplest case we just add the dummy as an explanatory variable. A simple model of hourly wage determination in which only gender and education matter:

$$wage = \beta_0 + \delta_0\, female + \beta_1\, educ + u$$

where female = 1 when the person is female and 0 when male.

Parameter	Interpretation
β_0	The intercept for males – the base group
$\beta_0 + \delta_0$	The intercept for females
δ_0	The difference in wage between females and males, given the same education (and the same u)
β_1	The return to education, assumed the same for both groups

Males are the **base group** or **benchmark group** – the group against which comparisons are made. The coefficient δ_0 is what determines whether there is discrimination: if $\delta_0 < 0$ then, for the same level of other factors, women earn less than men on average.

Graphically this is an intercept shift. The wage-education profiles for men and women are parallel, because the difference does not depend on the amount of education – that is precisely what an intercept dummy assumes.

2.2 How many dummies can you include?

> **Question**
>
> Why not also include a variable male, equal to one for men and zero for women?

Answer: only one of the two can be included. With two groups we need only two different intercepts, and the model already has β_0. Adding both dummies alongside the intercept introduces perfect collinearity, because:

$$female + male = 1$$

so male is an exact linear function of female and the constant. This violates MLR.3 and the model cannot be estimated. This is sometimes called the dummy variable trap.

The general rule: with g groups, include $g - 1$ dummy variables plus an intercept. The omitted group becomes the base group, and every coefficient is read relative to it.

2.3 Wage discrimination – a fuller model

Nothing much changes when more regressors are added. Taking males as the base group:

$$wage = \beta_0 + \delta_0\, female + \beta_1\, educ + \beta_2\, exper + \beta_3\, tenure + u$$

If education, experience and tenure are the relevant productivity characteristics, then:

$$H_0: \delta_0 = 0 \quad versus \quad H_1: \delta_0 < 0$$

Testing for discrimination requires nothing new – estimate by OLS and use the usual t-statistic. The result: taking a woman and a man with the same education, experience and tenure, the woman earns on average \$1.81 less per hour. The p-value is 0.00, so we reject H_0.

Note that the negative intercept here is not meaningful, because nobody in the sample has zero values of educ, exper and tenure simultaneously.

2.4 Dummy variables when the dependent variable is log(y)

A very common specification has log(y) on the left and dummies on the right. The coefficient then has a percentage interpretation: multiplied by 100 it is the approximate percentage difference in y, holding other factors fixed.

For a large coefficient the approximation is poor, and the exact percentage difference between $x_1 = 1$ and $x_1 = 0$ is:

$$100 \cdot [\, exp(\hat{\beta}_1) - 1 \,]$$

Applied to the wage equation

Re-estimating with log(wage) as the dependent variable gives $\hat{\delta}_0 = -0.301$:

Method	Calculation	Result
Approximate	100 × (−0.301)	−30.1%
Exact	100 · [exp(−0.301) − 1]	−26.0%

So the more accurate estimate is that a woman's wage is on average about 26% below a comparable man's – not 30%. As in Lecture 4, the approximation overstates the magnitude of a decrease.

2.5 Interactions among dummy variables

Dummies can be interacted with each other, just as quantitative variables can. Suppose we want to know how marital status affects the wage, allowing the marriage premium to differ by gender. Introduce married (= 1 if married) and the product female·married:

$$log(wage) = \beta_0 + \beta_1\, female + \beta_2\, married + \beta_3\, female{\cdot}married$$

$$+ \beta_4\, educ + \beta_5\, exper + \beta_6\, tenure + u$$

Two dummies interacted create four groups, and setting female = 0 and married = 0 eliminates all three dummy terms – so single men are the base group. Estimated on 526 observations:

Group	Expression	Approximate	Exact
Single men (base)	$\hat{\beta}_0 = 0.388$	–	–
Single women	$\hat{\beta}_1 = -0.097$	−9.7%	−9.2%
Married men	$\hat{\beta}_2 = +0.292$	+29.2%	+33.9%
Married women	$\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3 = -0.120$	−12.0%	−11.3%

Each figure is a proportional difference relative to a single man with the same education, experience and tenure. So the marriage premium for men is roughly 29% (more precisely 34%), while a married woman earns about 12% less than a single man. All coefficients are significant at the 1% level except the one on female.

> **Where the care is needed**
>
> No single coefficient gives the married-women differential. You must add three: $\beta_1 + \beta_2 + \beta_3$.
>
> The reason is that a married woman switches on female, married, and their interaction simultaneously. Reading β_1 alone would give the differential for a single woman.

2.6 Dummies for multiple categories – the same model, restated

The same four groups can be handled by defining a dummy for each non-base group directly, instead of using an interaction: 1) single men (a base group); 2) married men (*marmale*), 3) married women (*marfem*), 4) single women (*singfem*):

$$log(wage) = \beta_0 + \beta_1\, marmale + \beta_2\, marfem + \beta_3\, singfem + \ldots + u$$

The two specifications are statistically identical – same coefficients on the quantitative variables, same R-squared, same summary statistics. What changes is convenience of reading:

The one useful difference

In this version the coefficient on marfem is −0.120 – the married-women differential, read directly off a single coefficient with its own standard error and p-value.

That is exactly the sum $\hat{\beta}_1 + \hat{\beta}_2 + \hat{\beta}_3$ from the interaction model. Reparameterising has bought you a t-test on a quantity that previously required a Wald test.

Choose whichever parameterisation puts a standard error on the comparison you care about.

3. Slope dummy variables

So far dummies have shifted the intercept. Interacting a dummy with a quantitative variable instead allows the slope to differ across groups.

3.1 Allowing for different slopes

Suppose we want to test whether the return to education is the same for men and women, while still allowing a constant wage differential:

$$log(wage) = \beta_0 + \delta_0\, female + \beta_1\, educ + \delta_1\, female{\cdot}educ + u$$

Group	Intercept	Slope on education
Males (female = 0)	β_0	β_1
Females (female = 1)	$\beta_0 + \delta_0$	$\beta_1 + \delta_1$

So δ_0 measures the difference in intercepts between women and men, and δ_1 the difference in the return to education. Two cases are worth picturing:

- **$\delta_0 < 0$ and $\delta_1 < 0$:** women earn less at all levels of education, and the gap widens as education rises.
- **$\delta_0 < 0$ and $\delta_1 > 0$:** women earn less at low levels of education, but the gap narrows as education rises – and at some point a woman with the same education earns more.

3.2 Testing – and a trap

Adding experience and tenure and estimating, the two hypotheses of interest are:

Hypothesis	Meaning	p-value	Decision
H_0: $\delta_1 = 0$	The return to education is the same for men and women	0.59	Fail to reject
H_0: $\delta_0 = 0$ and $\delta_1 = 0$	Average wages are identical for men and women with the same education	0.00	Reject

The estimated return to education is 0.090 (9%) for men and 0.090 − 0.0072 = 0.083 (8.3%) for women. The difference of −0.72 percentage points is neither economically large nor statistically significant, so there is no evidence against equal returns to education.

The serious error to avoid

In this specification the coefficient on female, though still economically large, is no longer statistically significant. Together with the insignificant δ_1, it is tempting to conclude there is no evidence of lower pay for women.

That would be a serious mistake. Once female·educ is in the model, δ_0 measures the wage differential at educ = 0 – and very few people in the sample have anything close to zero education, so that differential is estimated very imprecisely.

The right test is the joint one, H_0: $\delta_0 = 0$ and $\delta_1 = 0$, using a Wald or redundant-variables test. It rejects decisively (p = 0.00). Wages for men and women are not identical at the same level of education.

3.3 Centring the interaction

The differential at zero years of education is not informative. Far more interesting is the gender differential at the average education level in the sample, about 12.6 years. Replace female·educ with female·(educ − 12.6) and re-estimate.

This changes only the coefficient on female and its standard error – everything else, including the fit, is unaffected. And now the coefficient on female is statistically significant: women earn about 30% less than men at the average level of education.

This is the same lesson as the centring discussion in Lecture 4, in a new setting: an insignificant coefficient in a model with interactions may simply mean the effect has been evaluated somewhere no data exist.

Model without centering

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 10:49
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.464712	0.122892	3.781465	0.0002
FEMALE	-0.210376	0.173969	-1.209273	0.2271
EDUC	0.090276	0.008715	10.35851	0.0000
FEMALE*EDUC	-0.007245	0.013562	-0.534164	0.5935
EXPER	0.004642	0.001628	2.850310	0.0045
TENURE	0.017436	0.002981	5.848873	0.0000

R-squared	0.392603	Mean dependent var	1.623268
Adjusted R-squared	0.386763	S.D. dependent var	0.531538
S.E. of regression	0.416245	Akaike info criterion	1.096254
Sum squared resid	90.09501	Schwarz criterion	1.144908
Log likelihood	-282.3149	Hannan-Quinn criter.	1.115304
F-statistic	67.22251	Durbin-Watson stat	1.774145
Prob(F-statistic)	0.000000		

Model with centering

Dependent Variable: LOG(WAGE)
Method: Least Squares
Date: 01/05/26 Time: 11:53
Sample: 1 526
Included observations: 526

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	0.464712	0.122892	3.781465	0.0002
FEMALE	-0.301388	0.037274	-8.085756	0.0000
EDUC	0.090276	0.008715	10.35851	0.0000
FEMALE*(EDUC-@MEAN(EDUC,"1 526"))	-0.007245	0.013562	-0.534164	0.5935
EXPER	0.004642	0.001628	2.850310	0.0045
TENURE	0.017436	0.002981	5.848873	0.0000

R-squared	0.392603	Mean dependent var	1.623268
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Sum squared resid	90.09501	Schwarz criterion	1.144908
Log likelihood	-282.3149	Hannan-Quinn criter.	1.115304
F-statistic	67.22251	Durbin-Watson stat	1.774145
Prob(F-statistic)	0.000000		

3.4 A richer example: race and baseball salaries

Estimated for 330 major league baseball players, with log(salary) as the dependent variable. Here black and hispan are indicators for the individual player (white players are the base group), percblck is the percentage of the team's city that is black, and perchisp the percentage Hispanic; other regressors measure productivity and longevity. The model includes the interactions black·percblck and hispan·perchisp.

Dependent Variable: LOG(SALARY)
Method: Least Squares
Date: 01/05/26 Time: 12:31
Sample: 1 353
Included observations: 330

Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	11.55329	0.130457	88.56011	0.0000
YEARS	0.066751	0.012707	5.253125	0.0000
GAMESYR	0.011073	0.002214	5.001141	0.0000
RBISYR	0.011830	0.003504	3.375798	0.0008
ALLSTAR	0.008788	0.002716	3.236138	0.0013
BLACK	-0.160710	0.122572	-1.311149	0.1907
HISPAN	-0.200180	0.152703	-1.310911	0.1908
BLACK*PERCBLCK	0.012754	0.004930	2.586868	0.0101
HISPAN*PERCHISP	0.020711	0.009779	2.117836	0.0350

R-squared	0.633437	Mean dependent var	13.51172
Adjusted R-squared	0.624301	S.D. dependent var	1.163087
S.E. of regression	0.712905	Akaike info criterion	2.187958
Sum squared resid	163.1431	Schwarz criterion	2.291569
Log likelihood	-352.0130	Hannan-Quinn criter.	2.229287
F-statistic	69.33771	Durbin-Watson stat	1.317453
Prob(F-statistic)	0.000000		

Redundant Variables Test
Equation: EQ05
Redundant variables: BLACK HISPAN BLACK*PERCBLCK HISPAN *PERCHISP
Specification: LOG(SALARY) C YEARS GAMESYR RBISYR ALLSTAR BLACK HISPAN BLACK*PERCBLCK HISPAN*PERCHISP
Null hypothesis: BLACK HISPAN BLACK*PERCBLCK HISPAN*PERCHISP are jointly insignificant

	Value	df	Probability
F-statistic	2.887124	(4, 321)	0.0226
Likelihood ratio	11.66371	4	0.0200

F-test summary:

	Sum of Sq.	df	Mean Squares
Test SSR	5.869339	4	1.467335
Restricted SSR	169.0125	325	0.520038
Unrestricted SSR	163.1431	321	0.508234

The four race variables are jointly significant at 5% – test that first, before interpreting any of them individually. Then:

Scenario	Calculation	Result
Black player, city with no blacks	coefficient on black = −0.161	16.1% less than a comparable white player
Black player, percblck = 15	−0.161 + 0.0128 × 15	about 3.1% more than whites
Hispanic differential vanishes at	0.200/0.0207	perchisp ≈ 9.67

Why this is not simply evidence of discrimination

The estimates imply that whites earn less than blacks and Hispanics in cities heavily populated by minorities. A pure discrimination story does not predict that.

City composition may matter through player preferences instead – perhaps the best black players locate disproportionately in cities with larger black populations.

The econometrics is sound; the causal interpretation is not settled by it. This is the Lecture 1 lesson resurfacing in a concrete case.

3.5 Summary of dummy variables as regressors

1. In the simplest case a dummy distinguishes two groups, and its coefficient estimates the ceteris paribus difference between them.
2. More than two groups is handled by a set of dummies: with g groups, include g − 1 dummies.
3. Dummies can be interacted with quantitative variables to allow slope differences across groups.
4. All dummy estimates are interpreted relative to the base group – the group for which no dummy is included.

4. Binary dependent variable: the linear probability model

Now the qualitative variable moves to the left-hand side. What if the event we want to explain is itself binary – employed or not, in the labour force or not, crisis or no crisis?

4.1 The model

Write the usual multiple regression with y taking only the values zero and one:

$$y = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k + u$$

Because y takes only two values, β_j cannot be read as the change in y from a one-unit increase in x_j – y either flips from 0 to 1, or from 1 to 0, or does not change. But the coefficients still have a useful meaning, and the reason is one clean fact:

$$P(y = 1 \mid x) = E(y \mid x)$$

For a zero-one variable the probability of "success" is the same as the expected value. So the regression function is the response probability:

$$P(y = 1 \mid x) = \beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k$$

This is the **linear probability model (LPM)** – so called because the response probability is linear in the parameters. Each slope measures the predicted change in the probability of success:

$$\Delta P(y = 1 \mid x) = \beta_j \Delta x_j$$

4.2 Example: female labour force participation

Let inlf = 1 if a married woman reported working for a wage outside the home at some point during 1975, and zero otherwise. Using data from Mroz (1987), where 428 of the 753 women in the sample were in the labour force, participation is modelled on husband's earnings (nwifeinc, $ thousands), education, past experience, age, number of children under six (kidslt6), and number aged 6–18 (kidsge6).

Dependent Variable: INLF
Method: Least Squares
Date: 07/27/26 Time: 11:51
Sample: 1 753
Included observations: 753

Variable	Coefficient	Std. Error	t-Statistic	Prob.
NWIFEINC	-0.003405	0.001448	-2.350841	0.0190
EDUC	0.037995	0.007376	5.151194	0.0000
EXPER	0.039492	0.005673	6.961866	0.0000
EXPER^2	-0.000596	0.000185	-3.226959	0.0013
AGE	-0.016091	0.002485	-6.476014	0.0000
KIDSLT6	-0.261810	0.033506	-7.813888	0.0000
KIDSGE6	0.013012	0.013196	0.986077	0.3244
C	0.585519	0.154178	3.797683	0.0002

R-squared	0.264216	Mean dependent var	0.568393
Adjusted R-squared	0.257303	S.D. dependent var	0.495630
S.E. of regression	0.427133	Akaike info criterion	1.147124
Sum squared resid	135.9197	Schwarz criterion	1.196251
Log likelihood	-423.8923	Hannan-Quinn criter.	1.166051
F-statistic	38.21795	Durbin-Watson stat	0.493840
Prob(F-statistic)	0.000000		

All variables except kidsge6 are statistically significant, and all significant variables have the signs economic reasoning would predict.

Interpretation	Figure
Another year of education raises the probability of participation by	0.038
Ten more years of education raises it by	0.038 × 10 = 0.38
Marginal effect of experience (entered as a quadratic)	0.039 − 2(0.0006) exper
Experience at which the effect vanishes	0.039/(2 × 0.0006) = 32.5 years

Notice that the 0.38 figure is a probability change of 38 percentage points – large, and a hint of the trouble in the next section: nothing in the model stops such increments from pushing a predicted probability past one.

4.3 Three problems with the LPM

Problem 1: heteroskedasticity is automatic

For a binary variable the conditional variance is determined by the conditional mean:

$$Var(y \mid x) = p(x)[1 - p(x)]$$

Since p(x) depends on the regressors, so does the variance. Heteroskedasticity is therefore present by construction – the only exception being the uninteresting case where the probability does not depend on any regressor.

This does not bias the OLS coefficient estimates, but the estimated variances are biased. The solution is robust standard errors (Lecture 6).

Problem 2: fitted probabilities can fall outside [0, 1]

Plug in certain combinations of regressor values and the predicted probability leaves the unit interval. What would it mean to predict that a woman is in the labour force with probability −0.10?

In this sample, of the 753 women, 16 fitted values are below zero and 17 are above one. A related and deeper problem is that a probability cannot be linearly related to a regressor over that regressor's whole range.

> **Predicting anyway: percent correctly predicted**
>
> Define $\tilde{y}_i = 1$ if $\hat{y}_i \geq 0.5$, and $\tilde{y}_i = 0$ if $\hat{y}_i < 0.5$. This works even when some fitted values lie outside [0, 1].
>
> Comparing $\tilde{y}_i$ with the actual y_i gives the frequencies of correct prediction for each outcome, and the overall proportion correct – the percent correctly predicted, a widely used goodness-of-fit measure for binary models.
>
> Here the LPM gets 73.4% correct.

Problem 3: the partial effect is constant

For any regressor entering in level form, the marginal effect is the same everywhere. The estimated effect of another year of education on the participation probability is always 0.038 – whether the woman has 5 years of schooling or 17.

That is implausible. In reality the effect might well be larger at 12 years than at 5, and it must eventually flatten, since the probability cannot exceed one. The fix is a non-linear model.

4.4 LPM: what to keep

1. The LPM is estimated by plain OLS and lets us explain a binary response with tools we already have.
2. The estimates are interpreted as changes in the probability of success given a one-unit increase in the regressor.
3. Its drawbacks: predicted probabilities outside [0, 1]; a constant marginal effect for level-form regressors; and built-in heteroskedasticity.

Despite all three, the LPM remains useful and is often applied in economics – not least because its coefficients are directly readable, which is exactly what the alternatives sacrifice.

5. Binary dependent variable: logit and probit

5.1 The class of models

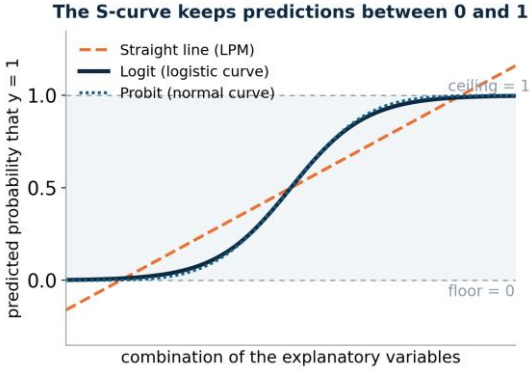
To keep probabilities inside the unit interval, pass the linear index through a function that cannot leave it:

$$P(y = 1 \mid x) = G(\beta_0 + \beta_1 x_1 + \ldots + \beta_k x_k)$$

where G takes values strictly between zero and one. Two choices account for the vast majority of applications:

Model	G(z)	Name of G
Logit	exp(z) / [1 + exp(z)]	The logistic function
Probit	$\Phi(z) = \int_{-\infty}^{z} \phi(v)\, dv$	The standard normal cumulative distribution function

Both are estimated by maximum likelihood rather than OLS. Plotted together, the two curves lie almost on top of one another – which is why the two models nearly always tell the same story.



> **The shape carries the economics**
>
> Near the flat ends of the curve, a change in x barely moves the probability. In the steep middle, the same change moves it a great deal.
>
> This is exactly what the LPM could not represent: the partial effect now diminishes as the probability approaches zero or one, which is both more plausible and what keeps the fitted values inside [0, 1].

5.2 Interpreting the output

In EViews, choose the BINARY estimation method and then Probit or Logit. The hardest part of these models is presenting and interpreting the results.

> **The one warning that matters most**
>
> Unlike the LPM, logit and probit coefficients are NOT changes in probability. Do not read their magnitudes directly.
>
> What you can read immediately is the sign: a positive coefficient always means "more likely", a negative one "less likely". Statistical significance is read from the p-value as usual.
>
> Relative magnitudes are also meaningful: for two continuous regressors the ratio of partial effects is β_j/β_h, and this ratio does not depend on x.

5.3 Getting from coefficients to probabilities

Because logit and probit coefficients are not probability changes, we need a further step to say anything quantitative. There are two standard ways of doing it, and they answer different questions.

Method 1: the partial effect at the mean (MEM)

1. Insert the mean values of all the explanatory variables into the estimated equation and compute the fitted probability.
2. Recalculate, increasing the variable under examination by one unit and leaving all others at their means.
3. The difference between the two is the impact of a one-unit increase for a person with average characteristics.

> **Handle dummy regressors with care**
>
> The mean of a dummy variable is a fraction – 0.43 "female", say – which corresponds to no actual person.
>
> When dummies appear in the equation, set the dummy first to one and then to zero and compare the two results, rather than evaluating at its mean. In practice this means computing the MEM separately for, say, men and women.

In EViews, for the labour force participation logit, with the equation object named eq_logit and the constant listed last:

1. Probability at the mean values of regressors

scalar ave_p_mean = @logit(eq_logit.c(1)*@mean(NWIFEINC) + eq_logit.c(2)*@mean(EDUC) + eq_logit.c(3)*@mean(EXPER) + eq_logit.c(4)*@mean(EXPER)^2 + eq_logit.c(5)*@mean(AGE) + eq_logit.c(6)*@mean(KIDSLT6) + eq_logit.c(7)*@mean(KIDSGE6) + eq_logit.c(8))

2. Recalculate, increasing the education by 1

scalar ave_p_educ = @logit(… eq_logit.c(2)*(@mean(EDUC)+1) …) – the same expression with education raised by one

3. Take the difference:

scalar mem_educ = ave_p_educ - ave_p_mean

Dependent Variable: INLF
Method: ML - Binary Logit (Newton-Raphson / Marquardt steps)
Date: 07/25/26 Time: 23:56
Sample: 1 753
Included observations: 753
Convergence achieved after 3 iterations
Coefficient covariance computed using observed Hessian

Variable	Coefficient	Std. Error	z-Statistic	Prob.
NWIFEINC	-0.021345	0.008421	-2.534620	0.0113
EDUC	0.221170	0.043440	5.091442	0.0000
EXPER	0.205870	0.032057	6.422001	0.0000
EXPERSQ	-0.003154	0.001016	-3.104093	0.0019
AGE	-0.088024	0.014573	-6.040232	0.0000
KIDSLT6	-1.443354	0.203585	-7.089692	0.0000
KIDSGE6	0.060112	0.074790	0.803749	0.4215
C	0.425452	0.860370	0.494499	0.6210

McFadden R-squared	0.219681	Mean dependent var	0.568393
S.D. dependent var	0.495630	S.E. of regression	0.425963
Akaike info criterion	1.088354	Sum squared resid	135.1762
Schwarz criterion	1.137481	Log likelihood	-401.7652
Hannan-Quinn criter.	1.107280	Deviance	803.5303
Restr. deviance	1029.746	Restr. log likelihood	-514.8732
LR statistic	226.2161	Avg. log likelihood	-0.533553
Prob(LR statistic)	0.000000		

Obs with Dep=0	325	Total obs	753
Obs with Dep=1	428		

Quantity	Value
Fitted probability at the means	0.631640
Fitted probability with education +1	0.681446
MEM for education	0.049807

So one extra year of schooling raises the predicted chance of labour force participation by about 5 percentage points – for a woman with average characteristics. For a probit model the procedure is identical with @cnorm in place of @logit.

> **Two things to state explicitly when you do this**
>
> The index numbering c(1) … c(8) follows the order in which the variables were listed when the equation was estimated. Here the constant was listed last, so c(8) is the intercept. Change the order of the equation specification and every index changes with it.
>
> Experience enters as a square. The code above uses mean of $(exper)^2$, i.e. the woman with average experience – not the mean of the squared series, which is a larger number because it also carries the variance of experience. Both conventions appear in practice and they give different answers, so say which one you are using.

The drawback of the MEM is easy to state: the "average person" may not represent anyone in particular. A woman with 12.3 years of education, 10.6 years of experience and 0.24 children under six does not exist.

Method 2: the average partial effect (APE)

The APE reverses the order of operations: compute an effect for every individual first, then average – instead of averaging the individuals first and computing one effect.

1. Predict at actual data. For every observation compute the fitted probability $\hat{p}_i^{o}$ using its own values of all regressors.
2. Predict with one variable shifted. Repeat, adding one unit to the variable under examination only; all other regressors stay at their actual values, giving $\hat{p}_i^{1}$.

3. Average the individual differences.

$$APE = (1/n)\,\Sigma\,(\hat{p}_i^{\,1} - \hat{p}_i^{\,0})$$

In EViews:

1. Forecast from the equation at actual values (Proc → Forecast), saving the series as inlf_f_actual.
2. Copy the education series so you can restore it, then: series educ = educ + 1
3. Forecast again with the adjusted education and actual values of the other regressors, saving as inlf_f_educ.
4. scalar ape_educ = @mean(inlf_f_educ - inlf_f_actual)
5. Restore the original education series.

The result is APE = 0.038983 – about 3.9 percentage points.

MEM versus APE: why they differ, and which to report

	MEM	APE
What is averaged	The regressors, before computing one effect	The effects, one per observation
Whom it describes	A hypothetical person with average characteristics	The actual sample
Value for education	0.0498 (5.0 p.p.)	0.0390 (3.9 p.p.)

Why the MEM is the larger number

The woman at the means has a fitted probability of about 0.63 – close to the steepest part of the logistic curve, where an extra year of education moves the probability a great deal.

The APE also counts the many women who are nearly certain to work, or nearly certain not to. For them the curve is flat and an extra year changes almost nothing. Averaging those in pulls the figure down.

Only the APE answers the policy question "what would happen on average if every woman got one more year of education". That is usually the number you want.

A useful consistency check: the LPM coefficient on education was 0.038, and the logit APE is 0.039. The linear probability model, for all its faults, delivers a good approximation to the average partial effect – which is exactly why it survives in applied work despite the objections in Section 4.3.

5.4 Comparing the three models

The estimates from LPM, logit and probit tell a consistent story: the same signs, the same variables significant. But the magnitudes are not directly comparable across model families. Rough conversion rules:

Comparison	Approximate factor	Reliability
Logit coefficient vs. LPM	× 4	Rough
Probit coefficient vs. LPM	× 2.5	Rough

Comparison	Approximate factor	Reliability
Logit coefficient vs. probit	× 1.6	The tightest of the three

Do not compare fit measures across families

The LPM reports an ordinary R-squared. Logit and probit report McFadden's pseudo-R-squared. These are different quantities.

Ranking models by reading one against the other is a mistake.

5.5 How good is the model?

For a binary outcome the natural check is how often the model predicts the right answer. Using a one-half cutoff, both logit and probit classify about 74% of the women correctly.

The benchmark to beat is the naive rule of guessing that everyone works, which is right 428/753 ≈ 57% of the time. Beating 57% with 74% is a real gain; a model that scored 58% would not be.

Logit and probit results are practically identical here, as they usually are. The choice between them rarely matters – use whichever your field prefers.

6. Summary

1. Qualitative information enters a regression through zero-one dummy variables, whose coefficients measure ceteris paribus differences relative to a base group.
2. With g groups include g − 1 dummies alongside the intercept; including all g creates perfect collinearity.
3. When the dependent variable is in logs, a dummy coefficient times 100 is an approximate percentage difference; $100 \cdot [\exp(\hat{\beta}) - 1]$ is exact.
4. Interacting two dummies creates four groups. Some group differentials require summing several coefficients – or reparameterising with a dummy per group so the comparison you want carries its own standard error.
5. Interacting a dummy with a quantitative variable allows different slopes. When you do this, the intercept dummy measures the gap at zero of the quantitative variable, which is often meaningless – centre it, and test jointly.
6. When the dependent variable is binary, OLS gives the linear probability model: simple and directly interpretable, but heteroskedastic, capable of predicting outside [0, 1], and constrained to constant marginal effects.
7. Logit and probit fix all three by passing the linear index through a cumulative distribution function. The cost is interpretability: coefficients are no longer probability changes.
8. To recover probabilities, compute either the partial effect at the mean (one hypothetical average person) or the average partial effect (averaged over the actual sample). The APE is normally the policy-relevant figure, and it lands close to the LPM coefficient.

The trade-off to carry away

The LPM is easy to interpret and wrong at the edges. Logit and probit are right at the edges and hard to interpret.

In practice, report the LPM for readable magnitudes and a logit or probit to show the conclusions do not depend on the linear approximation. If the three disagree about a sign, that is the finding worth investigating.

Key terms

Term	Meaning
Dummy (binary) variable	A variable taking only the values zero and one, encoding a qualitative characteristic.
Base (benchmark) group	The group for which no dummy is included; all dummy coefficients are read relative to it.
Dummy variable trap	Including a dummy for every group alongside an intercept, creating perfect collinearity.
Intercept dummy	A dummy entering on its own, shifting the intercept and leaving slopes unchanged.
Slope dummy	A dummy interacted with a quantitative regressor, allowing the slope to differ across groups.
Exact percentage difference	$100 \cdot [\exp(\hat{\beta}) - 1]$, used when a dummy coefficient in a log model is large.
Interaction of dummies	A product of two dummies; with two dummies the model describes four groups.
Linear probability model (LPM)	OLS applied to a binary dependent variable; coefficients are changes in $P(y = 1)$.
Response probability	$P(y = 1 \mid x)$. Equal to $E(y \mid x)$ when y is binary.
Percent correctly predicted	Proportion of observations for which the 0.5-cutoff prediction matches the actual outcome.
Logit	Binary response model with G the logistic function, $\exp(z)/[1 + \exp(z)]$.
Probit	Binary response model with G the standard normal cdf, $\Phi(z)$.
Maximum likelihood estimation	The estimation method used for logit and probit, in place of OLS.
Pseudo R-squared	McFadden's fit measure for logit and probit; not comparable with the OLS R-squared.
Partial effect at the mean (MEM)	The change in fitted probability from a one-unit increase, evaluated with all regressors at their means.
Average partial effect (APE)	The average across observations of each individual's change in fitted probability. Usually the quantity of policy interest.

Review questions

Use these to check your understanding before the seminar.

1. You define female = 1 for women. A colleague defines male = 1 for men instead. What changes in the estimated equation, and what does not?
2. Explain why a model with an intercept cannot include both female and male. Which assumption does it violate?
3. You have five regions and want region effects. How many dummies do you include, and how do you interpret each coefficient?
4. In log(wage) = … − 0.301 female + …, give both the approximate and the exact percentage differential. Why does the approximation overstate it?
5. In the four-group model with $\hat{\beta}_0 = 0.388$, $\hat{\beta}_1 = -0.097$, $\hat{\beta}_2 = 0.292$, $\hat{\beta}_3 = -0.316$, compute the differential for each of the four groups relative to single men. Which group is the base, and why?
6. Why does the married-women differential require summing three coefficients? Describe the reparameterisation that lets you read it off a single coefficient, and state what it buys you.
7. In log(wage) = $\beta_0 + \delta_0$female + $\beta_1$educ + $\delta_1$female·educ + u, write the intercept and slope for each gender. What does δ_1 measure?
8. The coefficient on female is insignificant and δ_1 is insignificant. Explain carefully why concluding "no evidence of a gender wage gap" would be a serious error, and state the correct test.
9. You replace female·educ with female·(educ − 12.6). Which coefficients change, which do not, and why does the coefficient on female become significant?
10. In the baseball example, a black player in a city with percblck = 15 earns more than a comparable white player. Does this refute discrimination? Give the alternative explanation and say what evidence would distinguish them.
11. Show that heteroskedasticity in the LPM is unavoidable. Does it bias the coefficients? What is the remedy?
12. In the participation model the effect of education is 0.038. Explain what is implausible about applying this figure to a woman with a predicted probability of 0.97.
13. The LPM predicts probabilities below zero for 16 women. Explain how you can still generate 0/1 predictions, and compute what the percent correctly predicted would need to exceed to be informative in this sample.
14. A logit coefficient is 0.80. Roughly what would you expect the corresponding probit and LPM coefficients to be? Which of the three conversions is most reliable?
15. Your logit has a pseudo-R^2 of 0.22 and your LPM an R^2 of 0.26. Can you conclude the LPM fits better?
16. Describe in words the difference between the MEM and the APE. Which averages first – the people or the effects?
17. For education, the MEM is 0.0498 and the APE is 0.0390. Explain why the MEM is larger, referring to the shape of the logistic curve.

18. A minister asks: "if every woman in the country completed one more year of schooling, how much would participation rise?" Which of the two figures answers this, and why is the other one wrong for this purpose?

19. Your equation includes a female dummy. Explain why computing the MEM at the mean of that dummy is unsatisfactory, and what you should do instead.

20. The LPM coefficient on education is 0.038 and the logit APE is 0.039. What does this near-agreement tell you, and does it excuse the LPM's three drawbacks?

References and further reading

Wooldridge, J. M. Introductory Econometrics: A Modern Approach – Chapters 7 and 17 cover the material of this lecture.

Asteriou, D., Hall, S.G. Applied Econometrics – Chapters 9 and 12.